

Chapter 122 — Bladder and Female Urethral Diverticula

BLADDER DIVERTICULA

- **Definition:** a subjectively large **herniation of bladder urothelium through the muscularis propria**, producing a thin-walled, urine-filled sac that communicates with the bladder lumen through a neck (ostium) of variable size.
- **Wall:** mucosa, subepithelial connective tissue (lamina propria), scattered thin muscle fibres and adventitia. An outer **fibrous pseudocapsule** is often present and gives a useful surgical plane at excision.
- The residual smooth muscle in the wall is **disorganised and non-functional**, so the sac cannot contract. Diverticula therefore **empty poorly at micturition**, leaving a large postvoid residual (PVR) urine volume — the origin of the characteristic presentation and imaging findings.
- **Neck anatomy (Fig. 122.1):** the diverticular neck is ringed by circular fibres of the middle detrusor layer (fine, circularly oriented fibres), with the longitudinal bundles of the outer layer running past it. This ring is the basis of the **narrow, sphincter-like neck** that transurethral neck resection later aims to disrupt.

Classification, Pathophysiology, and Etiology

- **About 90% of bladder diverticula occur in adults.** They are far commoner in males, with a **male-to-female ratio of about 9:1** in both adult and paediatric groups.
- Diverticula are classified as **congenital or acquired (secondary)**. Pathophysiology, presentation, clinical implications and imaging differ between the two.

CONGENITAL VERSUS ACQUIRED BLADDER DIVERTICULA

Feature	Congenital	Acquired (secondary)
Age	Childhood; peak under 10 years	Males usually after 60 years — the age of prostatic enlargement
Number	Usually solitary	Often multiple
Sex	Mostly male	Much commoner in males; uncommon in females and usually due to outlet obstruction
Site	Posterolateral to the ureteric orifice; about 90% near the ureterovesical junction (UVJ) ; often with vesicoureteral reflux (VUR)	Most commonly at the ureterovesical hiatus, but anywhere in the bladder
Cause	Congenital weakness of the detrusor , usually at the UVJ, with or without other lower urinary tract abnormality	Bladder outlet obstruction (BOO) or neurogenic vesicourethral dysfunction — one or the other is almost always present in adults
Bladder wall	Smooth-walled ; no significant trabeculation at cystoscopy	Associated with significant trabeculation
Size	Relatively larger	Relatively smaller
Commonest presentation	Acute urinary tract infection (UTI) from stasis	Incidental, during evaluation of lower urinary tract symptoms (LUTS), haematuria or UTI
Malignancy	Almost no increased association	Increased risk

Congenital diverticula

- May occur with **normal voiding dynamics and no BOO**, yet **up to 60%** are associated with an underlying syndrome, neuropathic voiding dysfunction or BOO. Incidence in children is about 1.7% in a large paediatric genitourinary database.
- **Less common presentations:** enuresis, pyelonephritis, acute retention and stones. **Secondary BOO** can arise when the diverticulum extends distally towards the bladder neck.
- In **prune-belly syndrome or posterior urethral valves (PUV)**, diverticula may lie at the **dome** with aberrant voiding dynamics or anatomy; these must be distinguished from the **urachal diverticula** seen in some paediatric conditions.
- **Syndromic associations:** Menkes syndrome (kinky hair, copper deficiency), Williams syndrome, Ehlers-Danlos syndrome and fetal alcohol syndrome. Genetic predisposition without a syndrome is unclear (reports in twins and a possible autosomal dominant family).
- Chromosomal testing has been suggested because of the syndromic links, but it is **not recommended in a child with a single simple lesion**.

Acquired diverticula

- **Associated factors in adults:** BOO from benign or malignant prostatic disease, bladder neck obstruction (hypertrophy, contracture) and urethral stricture — although obstruction is not demonstrable in every case.
- **About 70% are associated with benign prostatic hyperplasia (BPH).** Historically, moderate to large diverticula were found in about 1–6% of men with "prostatism".
- **In females**, careful evaluation of the outlet usually reveals an obstructive cause: dysfunctional voiding, vaginal prolapse, primary bladder neck obstruction (hypertrophy), urethral stricture, or **iatrogenic obstruction after anti-incontinence surgery** (Fig. 122.3).
- In **children and young adults**, acquired diverticula follow bladder neck obstruction (dysfunction), PUV or neurogenic vesicourethral dysfunction.
- **Hutch diverticulum:** a diverticulum **superolateral to the ureteric orifice, not involving the trigone**, in a **neuropathic bladder with VUR** — described in paraplegic patients; also seen in dysfunctional voiding.
- **Iatrogenic diverticula:** inadequate closure of the muscular layers after **cystotomy** leaves a weak point in the suture line; they also arise at the ureteroneocystostomy site after **ureteric reimplantation**.

Concept — A diverticulum is a mucosal hernia without functioning muscle, so it cannot empty; stasis, infection, stones, malignancy and the urodynamic artefacts all follow from that. Congenital diverticula are a weak wall under normal pressure; acquired diverticula are a normal wall under high pressure — which is why the acquired type always sends the search back to the outlet.

Diagnosis

Presentation and Evaluation

- **Acquired diverticula produce no specific symptoms;** most are found during investigation of **nonspecific LUTS, haematuria or infection**, or incidentally on imaging or endoscopy for these conditions. An enormous diverticulum may be found on imaging for unrelated complaints in a patient with no LUTS or UTI (Fig. 122.4).
- Symptoms, when present, come from **stasis** or **mass effect**. On direct questioning, incomplete emptying, lower abdominal fullness and **double voiding** may be attributable to a large diverticulum, but these overlap with prostatic enlargement and other outlet conditions.
- Congenital diverticula most often present with UTI; haematuria, abdominal pain or an abdominal mass may also occur.

- **Rare presentations:** lower-limb oedema from **iliac vein compression**; diverticula contained within **inguinal hernias**.
- **UTI in a male** is often what triggers the investigation that uncovers a diverticulum. Stasis both predisposes to UTI and makes an established infection **harder to eradicate**.

Initial evaluation (sequence): history and examination including **digital rectal examination** → prostate-specific antigen (PSA) in males where appropriate → quantify LUTS, query occult neurogenic causes (for example spinal surgery) and characterise prior lower urinary tract surgery → **urinalysis, urine culture and urine cytology**, especially if nonoperative management is contemplated.

- Patients should be counselled that **bladder cancer risk may be somewhat higher** with diverticula.
- Haematuria and pyuria are common. **Relapsing or persistent pyuria unresponsive to antibiotics** may by itself justify diverticulectomy in an otherwise asymptomatic patient.
- An **incidental diverticulum in an adult** should prompt evaluation for BOO or occult neuropathic dysfunction, plus endoscopy and imaging of both the lower and upper urinary tract.

Imaging

- Diagnosis rests on radiographic and endoscopic findings. **Cellules → saccules → diverticula** form a continuum of increasingly large, and therefore more severe, manifestations of **raised intravesical voiding pressure**. Cellules and saccules are small outpouchings between hypertrophied muscle bands (saccules larger). There are **no agreed size cut-offs**; the distinction is subjective.
- **Differential diagnosis of a fluid-filled paravesical structure:** müllerian cysts; uterine, ovarian and tubal abnormalities; urachal cysts; ectopic ureter or ureterocele; postsurgical change including lymphocele.

Voiding cystourethrography

- **Fluoroscopic voiding cystourethrography (VCUG)** is an **excellent detection study**, although false negatives occur. Anteroposterior, oblique and lateral views show anatomy, location and size; the study also detects **associated VUR** and, importantly, whether the diverticulum **empties with voiding**. PVR in the diverticulum and in the bladder should both be recorded.
- **VUR occurs in up to 13% of congenital diverticula.** Historically every such child underwent ureteroneocystostomy; the current approach is **selective reimplantation**, reserved for complicating factors such as renal scarring or progressive deterioration of renal function.
- **Paradoxical enlargement:** during a detrusor contraction, contrast flows from the high-pressure bladder through the ostium into the low-pressure diverticulum, which enlarges as the bladder empties. The bladder may discharge partly into the sac and partly through the urethra; after voiding stops, the empty bladder **refills from the diverticulum** (Fig. 122.2 — a fluid–fluid level forms as denser contrast settles below urine).

Cross-sectional imaging

- Characterises the **number, anatomy and location** of diverticula and looks for **masses within them**. Filling defects or other bladder abnormalities demand further investigation (Fig. 122.5: multiple filling defects proved urothelial carcinoma; a diffusely thickened, irregular wall with two diverticula from chronic BOO showed trabeculation only; asymmetrical wall thickening with the ostium visible).
- If the **neck is obstructed by tumour or otherwise non-patent**, contrast will not enter, and cross-sectional imaging may be required for diagnosis.
- Provides surgically relevant anatomy — ureters and adjacent structures such as the **rectum**, which large diverticula can distort.

Upper tract imaging

- Options: intravenous pyelography, ultrasonography, computed tomography (CT) or magnetic resonance imaging (MRI). In adults without haematuria or suspected malignancy, the goal is **silent hydroureteronephrosis** related to the diverticulum.
- **Adults versus children:** hydroureteronephrosis in **up to about 7% of adults**; upper tract abnormalities (scarring, dysplasia, hydronephrosis) in **up to 30% of children**.
- **Causes of hydronephrosis:** ureteric obstruction by the diverticulum; the underlying urodynamic abnormality that created it; associated VUR; inflammation; or an unrelated cause. A multichannel urodynamic study with VCUG, or a **videourodynamic study (VUDS)**, separates upper from lower tract causes (Fig. 122.6: hydronephrosis with an ipsilateral diverticulum and retention resolved after transurethral resection of the prostate; the diverticulum persisted but was smaller).
- **Ureteric deviation:** the pelvic ureter is most commonly deviated **medially**, occasionally laterally (Fig. 122.7). This guides the surgical approach and whether to place **preoperative ureteric stents**.
- A diverticulum that **encompasses the ureteric orifice** creates a functionally shortened intramural ureter and causes VUR.

Urodynamics

- In adults, BOO or neurogenic dysfunction causes the diverticulum, and **VUDS is very helpful**. The role of urodynamics in congenital diverticula is less well defined.
- **Failure to identify and treat the underlying urodynamic abnormality** before or during definitive surgery may lead to **recurrent diverticula** or other postoperative problems, especially in adults — so the evaluation is critical. Treating the abnormality may improve emptying enough to **avoid diverticulectomy**.
- **Associated findings:** BOO, impaired contractility, raised PVR and detrusor overactivity.
- **Pressure-sink effect:** as the detrusor contracts, bladder contents decompress into the diverticulum (the path of least resistance) rather than through the urethra, so **contractility appears falsely diminished**. When carefully measured, contractility does not differ from men with BPH without diverticula.
- **Raised PVR** may simply be urine retained in the diverticulum, regardless of contractility or obstruction.

Endoscopic Examination

Inspect the **entire interior of every diverticulum** for stones or abnormal epithelium (a **flexible cystoscope** is often needed) → record location relative to the ureters and outlet, and size (Fig. 122.8) → obtain **cytology directly from the diverticulum** → **biopsy any abnormal epithelium** with extreme care (Fig. 122.9).

- **Every lesion under consideration for biopsy is malignant until proven otherwise.**
- The wall is very thin because it lacks muscularis propria; **perforation during biopsy risks tumour seeding into the perivesical fat**, with adverse prognostic implications.
- A papillary urothelial tumour can be seen endoscopically inside a diverticulum whose CT shows no mass (Fig. 122.9) — imaging alone does not exclude tumour.
- Patients electing **nonoperative management** need close follow-up with **periodic endoscopy and cytology**. The natural history is unknown, and malignant transformation from stasis over time should be considered.

Concept — *The diagnostic workup has two jobs: find the cause (outlet or neurogenic) and exclude cancer inside a sac that imaging, endoscopy and even urodynamics each see imperfectly — so the tests are combined rather than any one relied upon.*

Associated Conditions

Bladder Outlet Obstruction

- **Acquired diverticula commonly coexist with BOO.** Common male causes: benign and malignant prostatic disease, urethral stricture. Less common: primary bladder neck obstruction (hypertrophy), vesicourethral anastomotic stricture after prostatectomy, and functional obstruction from neurogenic dysfunction (bladder neck or striated sphincter dyssynergia).
- **Treat the BOO before, or together with, the diverticulum.** Whether to do both at once or stage them is controversial.
- Especially with **small diverticula**, treating the BOO alone may permit complete, low-pressure emptying of the diverticulum and avoid future diverticulectomy.

Malignancy

- **Risk of malignancy is elevated in adults**, attributed to **urinary stasis** (prolonged carcinogen exposure) and **chronic inflammation**.
- **Inflammatory urothelial change** is found in **45–80%** of diverticulectomy specimens; overall **tumour prevalence within diverticula is 0.8–10%**. Unsuspected cancer is found on final pathology in a small but not insignificant number. Because diverticula are under-reported overall, the true tumour prevalence is probably considerably lower.
- **Histology: urothelial cell carcinoma (UCC) about 70–80%**, followed by **squamous cell carcinoma (SCC) 20–25%**.
- Tumours occur almost exclusively in adults, with a **peak at 65–75 years**.
- **Lack of muscularis propria** means these tumours are at greater risk of being **high grade and invasive**, with rapid transmural and **extravesical extension** and a poorer prognosis. It also risks early dissemination during biopsy or transurethral resection (TUR), and makes **precise pathological staging difficult**.
- **Prognosis** was historically poor — attributed to delayed diagnosis and advanced stage — but may be better in selected patients.
- Historically this justified **prophylactic excision of all diverticula**, including asymptomatic ones. It is now accepted that **observation with periodic endoscopic surveillance** is reasonable in asymptomatic patients without dysplasia, malignancy or complicating factors (for example upper tract deterioration).
- **Imaging** (CT, MRI, ultrasonography) identifies tumours within diverticula, but whether it reliably predicts extravesical invasion is unclear; one small series suggests tumour length and width on CT correlate with survival.
- **Clinical stage at presentation is the most important prognostic factor**: 5-year survival falls steeply from superficial to extradiverticular disease. Preoperative staging is nevertheless inaccurate, with **substantial upstaging on final pathology**.

TREATMENT STRATEGIES FOR TUMOURS IN BLADDER DIVERTICULA

Strategy	Position in the chapter
Aggressive individualised multimodal therapy (surgery, chemotherapy, radiotherapy)	May improve prognosis (small series)
Open exploration with partial or radical cystectomy without prior TUR	Advocated by some because post-TUR staging is difficult and often inaccurate
Individualised by clinical stage and pathological grade	Advocated by others

Strategy	Position in the chapter
Radical cystectomy	Same recurrence-free and overall survival as non-diverticular tumours when stratified by stage
Diverticulectomy alone	Can succeed for low-grade, low-stage tumours, but preoperative stage and grade are unreliable — only in select cases, with counselling and follow-up
Biopsy and fulguration ± intravesical chemotherapy or immunotherapy, with cystoscopic surveillance	Role undefined; TUR carries a higher perforation risk in these tumours

Concept — *The absent muscularis propria explains the whole malignancy section: it lets tumours reach the perivesical fat early, lets instruments perforate and seed, and removes the layer against which invasion is staged — so a diverticular tumour is treated as potentially invasive from the outset.*

Other Associated Conditions

- Many diverticula sit at the UVJ, explaining the high incidence of **ipsilateral ureteric abnormalities**. A ureter entering a diverticulum lacks muscular backing, giving a **functionally shortened intramural tunnel** (Fig. 122.7); there is a **very high prevalence of ipsilateral VUR** with congenital diverticula.
- Other considerations: **stones** within the diverticulum, **ureteric obstruction**, and rare but life-threatening **perforation or rupture**.

KEY POINTS: DIAGNOSIS OF BLADDER DIVERTICULA

- No defined, functional muscularis propria, so larger diverticula empty poorly at micturition.
- In adults, usually associated with BOO or neurogenic vesicourethral dysfunction; many are paraureteral.
- Most are diagnosed incidentally during evaluation of nonspecific LUTS or infection.
- Evaluate for BOO and, if present, treat it before or with definitive treatment of the diverticulum.
- Malignancy can arise within a diverticulum; consider early extravesical extension because muscularis propria is absent.

Management of Bladder Diverticula

- **Options:** observation, endoscopic management and surgical excision, chosen after weighing many factors (Fig. 122.10).

FIG. 122.10 ALGORITHM FOR THE EVALUATION AND THERAPY OF BLADDER DIVERTICULA

Branch	Sequence	End points
Entry	Known or suspected diverticulum → review history and LUTS → cystoscopy, cytology, VCUG → confirm diverticulum	—
Suspicious lesion	Biopsy → if cancer: determine type, grade and stage (metastatic evaluation) and image the upper tract; if benign: rejoin "no suspicious lesion"	Appropriate cancer therapy
No suspicious lesion — hydroureteronephrosis (HUN) on upper tract imaging	Urodynamics → if abnormal, treat the lower urinary tract abnormality → reassess. If unresolved HUN or normal urodynamics: decide whether the HUN is due to the diverticulum	Resolved HUN → surveillance; HUN not due to diverticulum → treat other causes; HUN due to diverticulum → assess surgical candidacy

Branch	Sequence	End points
No suspicious lesion — normal upper tract, asymptomatic and uncomplicated	Surveillance; if complications, symptoms or HUN develop → re-enter the algorithm	Surveillance
No suspicious lesion — normal upper tract, symptomatic or complicated	Urodynamics → if abnormal, treat it → reassess. Persistent symptoms or complications, or normal urodynamics → assess surgical candidacy	Resolved, uncomplicated → surveillance; good candidate → diverticulectomy; poor candidate → clean intermittent catheterisation (CIC)

Modified from Rovner ES, Wein AJ: Bladder diverticula in adults. In Resnick M, Elder JA, Spiranak JP, editors: Decision making in urology, ed 3, 2004.

- The algorithm is **cancer first, upper tract second, urodynamics third**: every branch passes through treatment of the underlying urodynamic abnormality before surgery is considered, and surgery is reserved for persistent symptoms, complications or HUN attributable to the diverticulum.

Observation and Nonsurgical Management

- Adults with **minimal symptoms and no complicating factors** may choose observation with surveillance, after counselling about the possibly increased malignancy risk, the need for reassessment, the unpredictable and potentially aggressive course of any malignancy, and the signs of progression — **increasing LUTS, haematuria and UTI**.
- **Surveillance schedule is not well defined** but consists of periodic reassessment of symptoms, urine studies including cytology, and endoscopy of the lower urinary tract.
- Patients with **poor emptying after relief of obstruction who remain symptomatic**, or who cannot or will not undergo excision, can be managed with **CIC or an indwelling catheter**; ongoing CIC with surveillance is acceptable if no complicating factors arise.
- **Children**: indications for diverticulectomy match adults, but with **multiple diverticula in a chromosomal syndrome**, observation and medical management are preferred — the connective tissue disorder impairs wound healing, raises perioperative risk and predisposes to recurrence.

Indications for Intervention

- Many incidentally found congenital or acquired diverticula are **managed expectantly** with periodic surveillance.
- Diverticula may be larger than the bladder (Fig. 122.11), but **size does not correlate with symptoms or complications** and is not an absolute indication for surgery.
- **Diverticulectomy is indicated for**: LUTS related to the diverticulum that do not respond to medical therapy; persistent symptoms; **chronic relapsing UTI**; **stones** within the diverticulum; **carcinoma or premalignant change**; and **upper tract deterioration from obstruction or VUR**.
- Because symptoms and complications arise from poor emptying and stasis, excision improves lower tract emptying **provided the primary problem (BOO) has been addressed**.
- Excision is usually **elective**; merits versus surveillance are discussed individually. The patient should be a reasonable surgical risk: correct or optimise nutritional, renal, cardiac and pulmonary factors and **treat preoperative UTI**. Prohibitive-risk patients should not undergo excision but may suit **CIC or endoscopic treatment**.

Staged Management for Bladder Outlet Obstruction

- **Concomitant** open, laparoscopic or robotic diverticulectomy with prostatectomy, or diverticulectomy with transurethral prostatectomy, is often performed; a **staged approach** is an alternative.
- If the diverticulum is attributable to BOO and there are no complicating factors, **treat the outlet first** → reassess the diverticulum → if emptying is satisfactory, symptoms resolve, and VUR, infection, malignancy and stones are absent, **ongoing surveillance may be all that is needed**.
- Emptying after relief of obstruction is assessed by VCUG, videourodynamics or cross-sectional imaging such as ultrasound.

Endoscopic Management

- **Candidates:** older or debilitated patients, poor candidates for open surgery, and patients undergoing **TUR of the prostate with an associated poorly draining diverticulum**.

TUR of the diverticular neck (sequence): incise the neck with the resectoscope loop or a **Collins knife** → carry incisions down to the bladder muscle fibres at the level of the ostium → circumferential resection of the entire neck if needed → the enlarged neck loses its sphincter-like narrowing and the diverticulum empties better during micturition.

- Generally safe and well tolerated, but it can cause **urinary retention through reversal of flow** — bladder contents "vent" into the diverticulum during voiding. It is **less successful in large diverticula**.
- **Neck TUR combined with fulguration of the entire urothelial lining** should obliterate the diverticulum or considerably reduce it. In a small short-term series it was reasonably effective for somewhat larger diverticula (≥ 4 cm) in patients unwilling or unable to undergo excision.

Operative Excision

- Open, laparoscopic or robotic, with or without concomitant treatment of BOO. The technique of diverticulectomy is covered in Chapter 124.

MANAGEMENT OPTIONS FOR BLADDER DIVERTICULA COMPARED

Option	Suitable for	Key caveat
Surveillance	Minimal symptoms, no complicating factors; after successful BOO treatment	Symptoms, cytology and endoscopy periodically; schedule undefined
CIC or indwelling catheter	Poor emptying after BOO relief, still symptomatic; unfit or unwilling for excision	Acceptable long term only without complicating factors
Endoscopic (neck TUR $\pm$ fulguration)	Elderly or debilitated; poor open candidates; at TURP with poorly draining diverticulum	Retention from venting; less effective in large diverticula
Diverticulectomy	Refractory LUTS, relapsing UTI, stones, carcinoma or premalignancy, upper tract deterioration	Outlet must be treated first or simultaneously

KEY POINTS: MANAGEMENT OF BLADDER DIVERTICULA

- Diverticulectomy is not indicated in all cases; reserve it for related symptoms, chronic relapsing UTI, stones, carcinoma or premalignant change, and upper tract deterioration from obstruction or VUR.
- Excision may be open or minimally invasive; treat concomitant BOO if present.

Concept — Management is sequenced around the cause: relieve the outlet first, then judge whether the sac still matters. Size never decides; complications, cancer and failure to empty do.

FEMALE URETHRAL DIVERTICULA

- **Definition:** female urethral diverticula (UD) are variably sized, **urine-filled periurethral cystic structures** adjacent to the urethra, **within the confines of the pelvic fascia**, connected to the urethra by an **ostium**. They are among the most challenging diagnostic and reconstructive problems in female urology.
- **Presentation ranges** from asymptomatic lesions noted incidentally on examination or imaging to debilitating, painful vaginal masses with incontinence, stones, severe dyspareunia or tumours. Anatomical variation in location, size and complexity makes **each case unique**.
- First described in 1805; the **modern era began in the 1950s with positive pressure urethrography (PPU)**, after which reported cases rose sharply — one mid-century series roughly doubled the number reported in the preceding 60 years. Ultrasonography and surface-coil or endoluminal MRI have since refined understanding (Fig. 122.12).

Anatomy of the Female Urethra

- A **musculofascial tube about 3–4 cm long**, from bladder neck to external meatus, suspended from the pelvic sidewall (tendinous arc of the obturator) by the **urethropelvic ligament** — two fused layers of pelvic fascia extending to the sidewall bilaterally (Fig. 122.13).
- The ligament has an **abdominal side (endopelvic fascia)** and a **vaginal side (periurethral fascia)**. **The urethra and most UD lie within and between these two leaves.**
- The urethra is a rich, vascular, **spongy cylinder** surrounded by an envelope of smooth muscle, skeletal muscle and fibroelastic tissue.
- **Periurethral glands:** tubuloalveolar glands in the thick vascular lamina propria (submucosa), posterolateral along the whole length but **most prominent over the distal two-thirds**, with most **draining into the distal third** (Fig. 122.14 — they drain distally but arborise proximally). **Most acquired UD are thought to arise from pathology of these glands.**
- **Skene (paraurethral) glands** are the largest and most distal periurethral glands; they drain **outside the lumen, lateral to the meatus.**
- **Muscle layers** (varying sphincteric function): inner longitudinal smooth muscle, outer circular smooth muscle, and a **skeletal muscle layer** that spans much of the length but is **most prominent in the middle third**. The skeletal layer is **U-shaped (horseshoe), incomplete on the vaginal aspect** where the urethra abuts the vagina.
- The **anterior vaginal wall** lies ventral to the urethra, separated from it by the periurethral fascia.
- Because the sphincters and UD overlap anatomically, **sphincter location and competence** matter when planning repair.
- **Innervation:** pudendal nerve (S2–S4); afferents travel in the pelvic splanchnic nerves.

PROXIMAL VERSUS DISTAL FEMALE URETHRA

Feature	Proximal	Distal
Lining	Urothelium	Nonkeratinised stratified squamous epithelium
Arterial supply	As for the adjacent bladder	Terminal branches of the inferior vesical artery via the vaginal artery along the superolateral vagina
Lymphatics	External and internal iliac nodes	Superficial and deep inguinal nodes

Feature	Proximal	Distal
Periurethral glands	Present but less prominent	Most prominent (distal two-thirds); most ducts drain into the distal third; Skene glands at the meatus

Concept — *The urethra lies in a fascial sandwich with its glands in the submucosa; a UD is gland disease expanding into the space between the fascial leaves, which is why preserving and reclosing the periurethral fascia governs the repair.*

URETHRAL DIVERTICULA

Pathophysiology and Etiology

- A UD is a **cavity dissecting within the fascia of the urethropelvic ligament**, usually an isolated cyst-like appendage with a **single discrete ostium** (Fig. 122.15). Complex patterns exist: extending partly around the urethra (**saddlebag**), anterior to the urethra, or **circumferential** (Fig. 122.16).
- **Congenital versus acquired**: UD in children may be a different entity from adult UD. Congenital cases in infants are reported, and some in young females regress spontaneously. Congenital anterior UD in **males** are a well-described but entirely different entity. Congenital Skene gland cysts are extremely rare. Paediatric diverticula have been attributed to an **ectopic ureter draining into a Gartner duct cyst** and to a forme fruste of **urethral duplication**.
- **The vast majority of UD are acquired and diagnosed in adult females**; two large series contained no patient under 10 years, arguing against a congenital cause. A congenital precursor that becomes symptomatic later remains unproven.

THEORIES OF FORMATION OF ACQUIRED URETHRAL DIVERTICULA

Theory	Mechanism and support	Evidence against or status
Childbirth trauma	Vaginal delivery herniates urethral mucosa through the muscle layers; forceps delivery reported to cause UD	Up to 20–30% of patients are nulliparous , which substantially discounts parity
Periurethral gland infection and obstruction (most widely accepted)	Infection → duct obstruction → suburethral abscess → rupture into lumen. Ostium posterolateral in the mid or distal urethra in ≥90% , matching gland location; periductal and interductal inflammation on Huffman's wax-model studies; strong link with Neisseria gonorrhoeae	About one-third of UD are proximal , where few ducts drain; periurethral abscesses should then be as common as UD but are rare; some regard UD as part of a continuum of periurethral cysts
High-tone, non-relaxing sphincter	Relative obstruction may contribute to some proximal UD	Interesting but unconfirmed
Iatrogenic mimics	Periurethral bulking agents (collagen and others) can simulate a UD radiographically; MRI helps distinguish them	Polypropylene mesh slings have no known association with UD formation

- **Huffman's anatomical work**: serial 10-µm sections and wax models showed the periurethral glands lie mainly **dorsolateral** to the urethra, arborise proximally, yet drain through ducts in the distal third.
- Initial infection and especially reinfection may come from **Escherichia coli**, other coliforms or vaginal flora.
- **Routh's concept** (now the most widely accepted theory): recurrent infection of the periurethral glands → obstruction → suburethral abscess → rupture into the urethral lumen → continual filling and pooling → stasis, reinfection and **epithelialisation of the cavity** → permanent UD. Reinfection, inflammation and recurrent **obstruction of the neck** produce symptoms and enlargement.

- **Noncommunicating UD** have been diagnosed on MRI; whether these are a forme fruste or UD with an obstructed ostium is unclear.

Young and Wahle's modern hypothesis (sequence): infection and obstruction of periurethral glands in the submucosa of the distal two-thirds → repeated infection and abscess formation → expanding mass displaces the spongy tissue → disrupts the muscular envelope → herniates into the periurethral fascia → dissects within the leaves of the periurethral fascia and urethropelvic ligament, **most commonly ventrally** (the classic anterior vaginal wall mass), sometimes laterally or dorsally → abscess ruptures into the lumen, creating the communication.

Concept — *The gland infection-abscess-rupture model predicts the posterolateral mid-to-distal ostium and the ventral vaginal mass; its weak point is the proximal UD, which is why alternative mechanisms are still entertained.*

Prevalence

- UD are found **in proportion to how hard they are sought**. Fewer than 100 cases were reported before 1950; diagnosis rose with PPU.
- **True prevalence is unknown**; reported **up to 1–6% of adult females** in some series. An accurate figure would require screening symptomatic and asymptomatic women in primary care, which has not been done.
- **Prevalence rises with symptom selection**: low in autopsy and asymptomatic populations, higher in incontinent women, and highest in highly symptomatic or tertiary referral cohorts (about 10% on endorectal-coil MRI of women with LUTS) — the latter do not reflect the general population.
- **Surgical and population-based rates are far lower** than the historical prevalence (population-based incidence under 0.02% per year), partly because UD may be asymptomatic, misdiagnosed or overlooked.
- **Race**: older series found Black women up to six times as likely to have UD, historically ascribed to urban academic referral bias. One large institutional series found most diverticulectomies in White women, yet national inpatient UD surgery was **threefold commoner in Black women**.

Diverticular Anatomy and Histology

- **Lining** may be urothelial, squamous, columnar, cuboidal or mixed; sometimes absent, the wall being fibrous tissue only. **Two-thirds show inflammation**.
- Histology is usually benign, but **about 10% of specimens show significant abnormality** — metaplasia, dysplasia or carcinoma — requiring long-term follow-up or further treatment.

CARCINOMA IN A URETHRAL DIVERTICULUM VERSUS PRIMARY URETHRAL CARCINOMA

Carcinoma arising in a UD	Primary female urethral carcinoma
Adenocarcinoma commonest, then UCC, then SCC	SCC commonest

- Fewer than 100 cases of carcinoma in UD have been reported. UD may be associated with **female urethral adenocarcinoma**; if so, observation or nonexcisional therapy (marsupialisation, endoscopic incision) should always be **combined with biopsy**.
- **No consensus on treatment**; recurrence is high with local treatment alone. Incidental malignancy found intraoperatively or on the final report is especially troubling.
- **No imaging modality** (ultrasonography, MRI) reliably and prospectively diagnoses a small malignancy in a UD.

- For cure, it is unclear whether **cystourethrectomy ± adjuvant external beam radiotherapy** beats **local excision + radiotherapy**. Chemotherapy alone after diverticulectomy (usually platinum-based), chemoradiation and targeted therapy can be considered.
- **Benign lesions within UD: nephrogenic adenoma** (pathologically difficult to distinguish from adenocarcinoma) and endometriosis.
- **Calculi occur in 4–10%**, from stasis or infection (Fig. 122.17 — a calcified density over the symphysis on the scout film proves to be a stone within a UD on voiding images). A stone **does not alter evaluation or surgical approach**; it is removed with the specimen.
- **Location**: within the periurethral fascia, bordered ventrally by the anterior vaginal wall. Size ranges from a few millimetres to several centimetres and **fluctuates** with inflammation, intermittent ostial obstruction and drainage.
- **Sagittal plane**: most often centred on the **middle third**, ostium **posterolateral**; may extend distally almost to the meatus or proximally beyond the bladder neck beneath the trigone (Fig. 122.18).
- **Axial plane**: may extend laterally, dorsally or circumferentially; may be bilobed (**dumbbell, saddlebag**) across the midline. Loculations are common; **about 6% have multiple UD**.
- Varying **sphincteric compromise** may exist because of the UD's position relative to the proximal and distal sphincter mechanisms, or from other factors — important in planning repair.

TABLE 122.1 DIVERTICULAR MORPHOLOGY AND CHARACTERISTICS (POOLED SUMMARY)

Characteristic	Pooled finding across contemporary series (873 patients)
Age	16–85 years; series means 36–52 years
Symptom duration	Series means 7–78 months (range 0–360)
Number	Single 294/361 (81%); multiple 67/361 (19%)
Configuration	Simple 282/386 (73%) ; saddlebag or horseshoe 65/386 (17%); circumferential 39/386 (10%)
Location	Proximal 64/261 (25%); mid-urethral 132/261 (51%) ; distal 65/261 (25%)
Size	Mean 21.1 mm; range 3–80 mm
Imaging sensitivity	MRI 97.2% ; double-balloon urethrography 94.7%; ultrasound 82.0%; VCUG 73.5%; cystoscopy 42.4%

- The table ranks the **diagnostic tests**: MRI and double-balloon urethrography detect almost all UD; ultrasound and VCUG miss about a fifth to a quarter; cystoscopy finds the ostium in fewer than half.
- **Pattern across the series**: MRI shows complexity better than VCUG; VCUG underestimates extent; the ostium is often invisible on every modality; MRI changed management in a minority; MRI sometimes found alternative diagnoses (vaginal cyst, endometriosis, ectopic ureter) and occasionally mislabelled a Skene gland abscess as UD; contrast-enhanced ultrasound localised ostia but needs urethral contrast injection; pseudodiverticula occurred after prior sling surgery.
- **Trap**: half of UD are mid-urethral, but a quarter are proximal — a proximal lesion is not excluded by the gland theory.

Presentation

- Most patients present between the **third and seventh decades**. Symptoms and signs are protean (Box 122.1).
- The **"three Ds"** — **dysuria, dyspareunia and (postvoid) dribbling** — are historical: only about **5% of contemporary cases have all three**, and individually or together they are neither sensitive nor specific.
- **Commonest symptoms**: **irritative LUTS** (frequency, urgency), **pain** and **infection**.

- **Multiple bouts of recurrent cystitis should raise suspicion of UD** (stasis in the sac).
- Other complaints: pain, vaginal mass, haematuria, vaginal discharge, obstructive symptoms, retention. Gentle compression of a tender or nontender anterior vaginal wall mass may express **urine or pus per meatus**. **Spontaneous rupture is extremely rare** but can produce a urethrovaginal fistula.
- **Size and complexity do not correlate with symptoms**: very large UD may be nearly silent while small, impalpable ones cause considerable distress. Symptoms **wax and wane**, perhaps with repeated infection and inflammation.
- **Misdiagnosis is common** — treatment for years for interstitial cystitis, recurrent cystitis, vulvodynia, endometriosis or vulvovestibulitis. Diagnostic delay runs to years across many physicians, even though about half have a palpable mass — underscoring a baseline level of suspicion and a thorough pelvic examination.

FREQUENCY OF PRESENTING FEATURES

Feature	Frequency
All three "Ds"	About 5%
Dyspareunia	12–24%
Postvoid dribbling	5–32%
Associated incontinence	About one-third
Recurrent cystitis or UTI	About one-third
Tender anterior vaginal wall mass at presentation	About one-third
Completely asymptomatic (incidental)	10–20%

BOX 122.1 SIGNS AND SYMPTOMS OF URETHRAL DIVERTICULA

Symptoms	Signs
Vaginal or pelvic mass; pelvic pain; urethral pain; dysuria	Recurrent UTI; haematuria
Frequency; urgency; hesitancy; postvoid dribbling; double voiding; sense of incomplete emptying	Vaginal or perineal tenderness; urinary retention
Dyspareunia; incontinence; vaginal or urethral discharge	Vaginal mass; urethral discharge on stripping the anterior vaginal wall

Pregnancy

- UD may present in pregnancy; whether pregnancy causes UD is unknown, but it may unmask symptoms.
- **Conservative management until after delivery** — antibiotics, aspiration or incision and drainage — avoids precipitating premature labour. Drainage during labour can facilitate delivery; vaginal delivery is possible; definitive repair follows delivery. Successful surgery during pregnancy has been reported, and **transvaginal aspiration** of symptomatic lesions is an option.

Concept — *Because UD symptoms are those of common lower urinary tract complaints and do not track lesion size, diagnosis depends on suspicion and examination rather than on any symptom pattern — the "three Ds" persist only as history.*

Evaluation and Diagnosis

- Diagnosis combines **history, physical examination, urinalysis and culture, cystourethroscopy and selected imaging**; urodynamics helps in selected cases.

History and Physical Examination

- Record LUTS, signs (for example haematuria), prior investigations and **response to previous therapy** — the basis for counselling about likely symptom improvement and for suspecting UD in diagnostic dilemmas.
- **Prior pelvic and anti-incontinence surgery:** bulking agents can appear as UD on imaging, and a **previous sling** can hinder access and repair.
- Document incontinence — presence, pad use and type (**urge vs stress vs postvoid dribbling**) — and **sexual function and dyspareunia**, since vaginal reconstructive surgery can worsen or improve them.
- **Palpate the anterior vaginal wall** for masses and tenderness; record location, size and consistency. Most UD lie **ventrally over the mid and proximal urethra**, i.e. **1–3 cm inside the introitus** (Fig. 122.19).
- **Lesions near or distorting the meatus** are more likely Skene gland cysts or abscesses; distinguishing them may require **MRI**.
- UD may lie anterior to the urethra, extend partly or fully around it, or extend **proximally to the bladder neck**, distorting the outlet and trigone. Such lesions raise the risk of intraoperative **bladder and ureteric injury** and of postoperative voiding dysfunction and incontinence.
- **A particularly hard mass suggests a calculus or malignancy** and mandates further investigation.
- **Stripping ("milking") the anterior vaginal wall** distally to express pus or urine is often described, but it fails to produce meatal discharge in the majority.
- Assess vaginal **atrophy, rugation and elasticity**: atrophic, poorly oestrogenised tissue is used for flaps, so **preoperative topical oestrogen** may improve it. Assess introital capacity — a **narrow introitus may require episiotomy**. Perform a **provocative stress test** and assess for pelvic organ prolapse.

Urine Studies

- **Urinalysis and culture are mandatory.** Commonest organism **E. coli**; other gram-negative enterics, *N. gonorrhoeae*, *Chlamydia*, streptococci and staphylococci are also found.
- **A sterile culture does not exclude infection** — many patients are on antibiotics at presentation. Cytology if irritative LUTS or suspected malignancy.

Cystourethroscopy

- Aims: **visualise the ostium** and exclude other causes of LUTS.
- A **flexible cystoscope** (no beak) or **rigid female cystoscope** (short beak) keeps the irrigant outflow beside the lens, distending the short female urethra for better luminal views.
- **Manoeuvre:** compress the bladder neck while an assistant presses on the diverticular sac, or compress the UD digitally during urethroscopy — pus is often seen issuing from the ostium.
- **Ostium usually posterolateral at 4 and 8 o'clock at the mid-urethra**, but often very hard to find (Fig. 122.20); success ranges **15–89%**.
- Highly symptomatic patients often need **anaesthesia** for endoscopy.
- A positive study helps planning; **failure to find an ostium must not deter further investigation or repair**.

Urodynamics

- Useful for patients with **incontinence or significant voiding dysfunction**, including documenting stress urinary incontinence (SUI) before repair.
- **About one-third present with incontinence; up to 50% demonstrate SUI** on urodynamics.
- **VUDS is optimal:** it combines VCUG and urodynamics, consolidating evaluation and reducing catheterisations. It distinguishes **genuine SUI** (needs an additional anti-incontinence procedure) from **paradoxic dribbling from the UD** (cured by diverticulectomy alone).

- **Urethral pressure profilometry** may show a **biphasic pattern or pressure drop** at the level of the lesion.
- A few patients show **BOO** from the obstructive or mass effect of the UD; SUI may coexist with obstruction, and both can be treated by a carefully planned operation.

Imaging

- High-quality preoperative imaging must both **diagnose** and **map the anatomy** — especially the relationship to the **proximal urethra and bladder neck**.

Diagnostic Contrast Radiography

- **No single study is the gold standard.** Choice depends on availability, cost and radiologist expertise.
Options: double-balloon PPU, VCUG, intravenous urography, ultrasonography and MRI with or without an endoluminal coil.

Double-balloon PPU (historically optimal; sequence): specialised **Trattner catheter** with two balloons several centimetres apart → one balloon at the external meatus, one at the bladder neck → both inflated to seal the urethral lumen → contrast infused under slight pressure through a side hole between the balloons (Fig. 122.21) → urethra distends and contrast is forced into the UD. Outstanding images and, unlike VCUG, **independent of voiding**; but rarely performed and the catheter is not widely available.

- **VCUG** (Fig. 122.22): widely available and familiar; **sensitivity 44–95%**. Patients often cannot void in the radiology suite (catheter pain, psychogenic inhibition), and without voiding the UD is often not seen — **a VCUG without voiding or postvoid images that shows no UD is nondiagnostic**. If voiding fails under fluoroscopy, the patient voids in private and a **postvoid film** is taken. Poor flow underfills the UD and **underestimates size and complexity** (Fig. 122.23).
- **Three-dimensional CT VCUG** with reconstructions is novel and not yet widely used.
- **Intravenous urography or CT urography**: when the upper tract must be delineated or a **congenital ectopic ureter** is suspected as the cause of an anterior vaginal wall mass; the postvoid film occasionally shows the UD.

Ultrasonography

- Abdominal, transvaginal, translabial and transurethral approaches; **transvaginal imaging** often gives size and location. The UD is **anechoic or hypoechoic with enhanced through-transmission**.
- Advantages: relatively noninvasive, **no radiation, no need to void**. Limitation: may not give detailed high-resolution surgical anatomy.
- **Contrast-enhanced sonourethrography** (transvaginal or transurethral) improves accuracy but needs a catheter — as invasive as VCUG — and a skilled sonographer.

Magnetic Resonance Imaging

- Relatively noninvasive, high-resolution, multiplanar. **UD are low signal on T1 and high signal on T2**. Distinct advantages over VCUG: **wholly independent of voiding** and **free of ionising radiation**.
- **Surface-coil** (Fig. 122.24) and **endoluminal MRI (eMRI)** techniques exist. eMRI places the coil in a body cavity next to the area of interest — **intravaginal or intrarectal** — improving signal-to-noise ratio and resolution, with excellent sensitivity and specificity.
- **Surface-coil MRI and eMRI appear superior to VCUG and/or PPU**, but are expensive and not universally available. Against PPU, MRI agrees well on the diagnosis, less well on complexity and location, and least well on demonstrating the urethral connection.
- **Pitfalls**: several lesions, including **periurethral bulking agents**, can be misdiagnosed as UD, and MRI may be **inadequate for detecting malignancy** within a UD.
- **Contraindications** are few: metallic foreign bodies, claustrophobia, intolerance of the endoluminal probe.

MRI FEATURES: BULKING AGENT VERSUS URETHRAL DIVERTICULUM

Feature	Favours bulking agent	Favours UD
Number	Multiple lesions	Usually single
Location	Upper or upper-mid urethra	Mid-urethra, posterolateral
Contents	No internal fluid or fluid–fluid level	Fluid
Urethral connection	None	Ostium
Signal	T1 isointense and T2 hyperintense to muscle, but lower T2 signal than urine	Low T1, high T2 (urine-like)

IMAGING MODALITIES FOR URETHRAL DIVERTICULA COMPARED

Modality	Needs voiding?	Radiation	Strength	Limitation
Double-balloon PPU	No	Yes	Outstanding urethral images	Trattner catheter scarce; rarely performed
VCUG	Yes	Yes	Widely available; familiar	Nondiagnostic without voiding; underestimates size
IVU / CT urogram	No	Yes	Upper tract; ectopic ureter	Only occasionally shows UD
Ultrasound	No	No	Noninvasive; size and location	Less surgical detail; contrast version needs catheter
MRI (surface coil, eMRI)	No	No	Best overall; complexity and extent	Cost, availability; mimics; misses small cancers

Differential Diagnosis

- Periurethral masses form a wide spectrum; many can be diagnosed on **history and examination alone**, others need judicious imaging and cystoscopy to exclude UD.

PERIURETHRAL MASSES COMPARED

Lesion	Origin and site	Distinguishing features	Treatment
Urethral diverticulum	Periurethral glands; mid-proximal urethra	Communicates with lumen; meatus not deformed	Excision and reconstruction
Vaginal leiomyoma	Smooth muscle of vaginal wall; anterior wall	Smooth, firm, round, freely mobile, nontender ; oestrogen dependent	Transvaginal excision or enucleation
Skene gland cyst or abscess	Paraurethral glands; lateral or inferolateral to meatus	Distal; distorts the meatus ; classically noncommunicating	Aspiration, marsupialisation, incision and drainage, excision (curative)
Gartner duct cyst	Mesonephric remnant; anterolateral vaginal wall, cervix to introitus	May drain an ectopic upper-pole ureter	Observe, sclerotherapy, excision or marsupialisation
Vaginal wall cyst	Mesonephric, müllerian, endometriotic, urothelial or epidermoid	Small, asymptomatic; type known only on histology	Simple excision if symptomatic
Urethral mucosal prolapse	Circumferential mucosal eversion at meatus	Beefy red "doughnut" around the whole meatus	Topical creams, sitz baths; circumferential excision
Urethral caruncle	Inflammatory lesion of distal urethra	Reddish exophytic mucosa-covered meatal mass; postmenopausal	Topical oestrogen or anti-inflammatory; excision if large or refractory

Lesion	Origin and site	Distinguishing features	Treatment
Bulking agent	Prior injection for SUI	History; noncommunicating; cystic on imaging	Transvaginal excision if symptomatic

Vaginal leiomyoma

- Benign mesenchymal tumour of vaginal smooth muscle (Fig. 122.25); uncommon (about 300 cases reported) and a small minority of periurethral masses; may be **misdiagnosed as UD**.
- Symptoms relate to size — mass effect, obstruction, pain, dyspareunia; **fourth to fifth decade**.
- Like uterine leiomyomas, **oestrogen dependent and regress at menopause**. Transvaginal excision or enucleation is often curative and is recommended to **confirm the diagnosis, exclude malignancy** and relieve symptoms.

Skene gland abnormalities

- Cysts and abscesses are similar lesions differentiated clinically (Fig. 122.26): small cystic masses just **lateral or inferolateral to the meatus**, lined by transitional or stratified squamous epithelium. **Abscesses** are very tender and inflamed, and pus may be expressed from the ductal orifice.
- **Classically, unlike UD, they do not communicate with the urethral lumen.**
- Seen in **neonates** and young to middle-aged women; symptoms include dysuria, dyspareunia, obstruction and pain.
- **Clinical distinction:** Skene lesions are distal and often **distort the meatus**; UD are usually mid or proximal and do not.
- Treatments: aspiration, marsupialisation, incision and drainage, simple excision — **excision is curative**.
- **Skene gland adenocarcinoma:** because of **homology with the prostate**, PSA may be raised and normalises with treatment.

Gartner duct abnormalities

- **Mesonephric remnants** on the **anterolateral vaginal wall** from cervix to introitus. Being mesonephric, they may receive **ectopic ureters** from poorly functioning or nonfunctioning **upper-pole moieties of duplex systems** (Fig. 122.27 — a turgid, debris-filled ectopic ureter on the anterior vaginal wall in a woman with lifelong incontinence and recurrent pyelonephritis); single-system ectopia is much less common in females.
- The proportion with ipsilateral renal abnormality is unclear, so **upper tract evaluation is recommended**. About **6% of patients with unilateral renal agenesis have a Gartner duct cyst**; **up to 50% with Gartner duct cyst and renal dysplasia** have ipsilateral müllerian duct obstruction.
- **Treatment:** asymptomatic with a nonfunctioning moiety → observe; aspiration with sclerotherapy has succeeded; symptomatic → simple excision or marsupialisation; **functioning renal moiety → individualise**.

Vaginal wall cysts

- Usually small, asymptomatic anterior wall masses that may enlarge to cause LUTS or dyspareunia (Fig. 122.28 — misdiagnosed as a cystocele for years).
- Origins: mesonephric (Gartner), paramesonephric (müllerian), endometriotic, urothelial or epidermoid (inclusion). **Specific diagnosis only on histology**; subtype is usually of little consequence, but **epidermoid cysts follow trauma or vaginal surgery**. **Müllerian cysts are commonest**, then epidermoid, then mesonephric; the remainder are Bartholin-type, endometriotic or indeterminate.
- Must be distinguished from UD; **simple excision** if symptomatic.

Urethral mucosal prolapse

- Circumferential herniation or eversion of urethral mucosa at the meatus: a **beefy red, doughnut-shaped lesion completely surrounding the meatus**. May be asymptomatic or cause bleeding, spotting, pain or urinary symptoms.
- **Two populations: prepubertal and postmenopausal females**. A predilection for young Black girls is not confirmed by recent series.
- **Children**: often related to **Valsalva voiding or constipation**, with eversion through a pathologically loose attachment between the urethral smooth muscle layers. **Postmenopausal**: causation less clear, epidemiologically linked to **oestrogen deficiency**.
- **Medical**: topical creams (oestrogen, anti-inflammatory) and/or sitz baths. **Surgical**: cauterisation, ligation around a Foley catheter, or **complete circumferential excision** with suture reapproximation of the remaining mucosa to the vaginal wall (few complications).
- **Bottom line**: medical treatment often fails or recurs; surgical excision gives **superior results** to medical management or catheter ligation, with urethral stenosis a possible complication.

Urethral caruncle

- Inflammatory lesion of the distal urethra, usually in **postmenopausal** women: a reddish, **exophytic, mucosa-covered mass at the meatus**. Often asymptomatic and incidental; when irritated causes underwear spotting or pain, less often voiding symptoms; rarely **thromboses** into a discoloured mass (Fig. 122.29 — simple excision curative).
- **Aetiologically related to mucosal prolapse**; chronic irritation produces haemorrhage, necrosis and inflammatory growth, matching the histology.
- **Atypical appearance or behaviour** → **excise** to exclude coexisting or mimicking disease: intestinal metaplasia, tuberculosis, melanoma, lymphoma.
- Evidence is sparse; initial **conservative** treatment with topical oestrogen or anti-inflammatory creams and sitz baths.

Simple excision of a large or refractory caruncle (sequence): grasp the tip → apply traction to expose the base fully → excise → reapproximate the mucosa with absorbable suture if the defect is large → Foley catheter for several days, around which the mucosa usually heals.

URETHRAL MUCOSAL PROLAPSE VERSUS URETHRAL CARUNCLE

Feature	Mucosal prolapse	Caruncle
Shape	Circumferential doughnut around the whole meatus	Exophytic mass at the meatus
Population	Prepubertal and postmenopausal	Postmenopausal
Cause	Valsalva, constipation, loose muscle layers; oestrogen deficiency	Related to prolapse; chronic irritation
Definitive treatment	Circumferential excision	Simple excision

Periurethral bulking agents

- Transurethral or periurethral injection for SUI can produce an anterior vaginal wall mass that **appears cystic on imaging and is often misdiagnosed as UD**; a careful history reveals it.
- Usually **noncommunicating and largely asymptomatic**; symptomatic lesions can be excised transvaginally.

Concept — Two questions sort almost every periurethral mass: *does it communicate with the urethra, and where does it sit relative to the meatus?* Only UD communicate; lesions at or distorting the meatus are Skene, prolapse or caruncle; firm mobile solid masses are leiomyomas.

Classification of Urethral Diverticula

CLASSIFICATION SYSTEMS FOR URETHRAL DIVERTICULA COMPARED

System	Basis	Use and status
L/N/S/C3 (Leach)	Location, number, size, configuration, communication site, continence status — analogous to cancer staging	Standardises description; not prospectively applied or validated ; not widely adopted
Ginsburg and Genadry	Location determines surgical approach	Distal → marsupialisation ; proximal → excision and reconstruction
Leng and McGuire	Presence or absence of a preserved periurethral fascial layer	Deficient fascia (after prior vaginal or urethral surgery) = pseudodiverticulum → needs additional reconstruction with a tissue flap or graft

Urethral Diverticula and Stress Urinary Incontinence

- SUI may coexist with UD. **Significant de novo SUI occurs in 7–16%** after diverticulectomy without a concomitant anti-incontinence procedure. When minor leakage is counted, de novo SUI approaches half, but most is minor and few need further surgery.
- **Risk factors for de novo SUI:** size ≥ 30 mm, **proximal location**, **circumferential configuration** and **shorter urethral length**. A complex circumferential configuration on MRI carried more postoperative SUI than a simple UD in one series, but this has not been found consistently.
- **Not every UD patient needs simultaneous SUI surgery:** extra time, cost and morbidity must be weighed against convenience. **About 50% of SUI improves or resolves after diverticulectomy alone**, so routine concomitant surgery risks overtreatment.
- **Bothersome symptomatic SUI → a concomitant anti-incontinence procedure can be offered** (Fig. 122.30); same-setting repair is well described. **No or mild non-bothersome SUI → diverticulectomy alone**, then reassess after recovery.
- **Synthetic material (mid-urethral polypropylene mesh) must not be used synchronously with UD surgery** — increased risk of urethral erosion and infection. Whether it is safe months or years later is unknown.
- **Autologous pubovaginal fascial slings** give satisfactory results as a concomitant procedure.
- **Staged approach:** most persistent or de novo SUI after transvaginal UD excision resolves with 12 months of conservative measures.

Concept — *The tissue dissected at diverticulectomy is the continence mechanism, so surgery can unmask, create or cure SUI. Hence only bothersome SUI is treated at the same sitting, and only with autologous tissue next to a freshly reconstructed urethra.*

KEY POINTS: DIAGNOSIS OF FEMALE URETHRAL DIVERTICULA

- Often diagnosed during evaluation of nonspecific LUTS, dysuria, pelvic pain, vaginal mass, dyspareunia and recurrent UTI.
- PPU, VCUG, ultrasonography and MRI may all be used; each has advantages and disadvantages.
- The differential of periurethral vaginal masses is extensive; careful history, examination and appropriate studies usually give a definitive preoperative diagnosis.

Surgical Repair of Female Urethral Diverticula

Indications for Repair

- **Not all UD mandate excision.** Some are asymptomatic or incidental; some patients are unwilling or medically unfit. The **natural history is unknown** — whether untreated UD progress in size, symptoms or complexity is not known. Surgical risks include **de novo incontinence, urethral stricture and vaginal scarring**.
- However, **carcinomas arising in UD are repeatedly reported**, some asymptomatic and not identified on imaging — so patients choosing **nonoperative management must be counselled**.
- **Nonoperative management: low-dose antibacterial suppression and digital stripping of the anterior vaginal wall after micturition** to prevent dribbling and reduce stasis-related UTI. Whether long-term cancer surveillance is needed is unknown.
- **Symptomatic patients** — dysuria, refractory bothersome postvoid dribbling, recurrent UTI, dyspareunia, pelvic pain — may be offered excision.

FIG. 122.30 ALGORITHM FOR TREATMENT OF URETHRAL DIVERTICULA

Stage	Pathway
Suspected UD	History, examination, urinalysis and cytology + imaging (VCUG, PPU, MRI, ultrasound) + cystoscopy
UD not found	Treat other conditions as appropriate
Diagnosis confirmed — asymptomatic, poor medical condition or not wanting surgery	Suppressive antimicrobial therapy + postvoid stripping of the urethra ± long-term surveillance
Symptomatic, complicating factor (for example stones) or elects repair — no LUTS	Transvaginal repair of UD
Symptomatic with LUTS	Urodynamics → SUI confirmed: transvaginal repair with anti-incontinence surgery ; no SUI: treat other urodynamic findings or LUTS, then transvaginal repair

Techniques for Repair

Alternative techniques

- Since the original 1805 description (transvaginal incision and packing with lint): **transurethral and open marsupialisation**, endoscopic unroofing, fulguration, incision and obliteration with oxidised cellulose or polytetrafluoroethylene, coagulation, and excision with reconstruction — the last now most common.
- **Transvaginal marsupialisation (Spence-Duckett)** of distal lesions may reduce operative time, blood loss and recurrence.
- **Risks:** extending the incision too far proximally causes **vaginal voiding** or damages the proximal and distal sphincter mechanisms (**SUI**); in sexually active women a **pseudoseptum** may form between urethra and anterior vaginal wall, causing **dyspareunia**.
- Therefore applicable only to **UD of the distal third in women who are not sexually active**, in very select cases; not commonly performed.

Excision and reconstruction

- The **commonest modern approach**; principles are set out in Box 122.2. Minor areas of disagreement: vaginal incision (**midline vs inverted U vs inverted T**), whether the entire mucosalised portion must be removed, and drainage (**urethral catheter alone vs urethral plus suprapubic**).

- **Complex reconstructions: bipediced vaginal wall flap** for UD with urethrovaginal fistula; laterally based vaginal flaps; various approaches for anterior or circumferential lesions; **laparoscopic or robotic** approaches for anterior (dorsal) and proximal UD.
- A systematic review found **insufficient data to distinguish outcomes or complications** between techniques. The technique below follows Leach.

BOX 122.2 PRINCIPLES OF TRANSVAGINAL URETHRAL DIVERTICULECTOMY

No.	Principle
1	Mobilisation of a well-vascularised anterior vaginal wall flap(s)
2	Preservation of the periurethral fascia as a separate layer
3	Identification and excision of the neck or ostium
4	Removal of the entire UD wall or sac (epithelium)
5	Watertight urethral closure
6	Multilayered, nonoverlapping closure with absorbable suture
7	Closure of dead space
8	Preservation or creation of continence

Preoperative preparation

- **Urine must be sterile**; do not operate in acute UTI. Prophylactic antibiotics help ensure sterility; **preoperative parenteral antibiotics**, especially with recurrent or persistent UTI.
- **Postvoid stripping** of the anterior vaginal wall empties the UD and prevents stasis and UTI — not possible with noncommunicating or painful UD.
- **Topical oestrogen for several weeks** in postmenopausal atrophic vaginitis improves tissue for dissection and flap mobilisation.
- **Counselling is paramount**: pain, dyspareunia, voiding dysfunction, UTI and incontinence may persist despite technically successful surgery.
- Symptomatic SUI → offer simultaneous anti-incontinence surgery. **Preoperative VUDS** maps the UD, assesses bladder neck competence, confirms SUI, and separates true SUI from postvoid dribbling, vaginal voiding and paradoxical incontinence from a urine-filled UD.

Procedure (Fig. 122.31)

1. **Position and exposure**: dorsal lithotomy, pressure points padded, adjustable padded stirrups; vaginal antiseptic preparation; weighted vaginal speculum and Scott (ring) retractor with hooks, Allis clamps. A **posterolateral episiotomy** occasionally helps, but the mid-urethral location of most UD usually makes it unnecessary.
2. **Drainage**: urethral Foley catheter. If wanted, place a **suprapubic tube at the start** (Lowsley tractor, or percutaneously under cystoscopic guidance) — **never at the end**, when it would traverse and risk disrupting the fresh urethral suture line.
3. **Incision**: mark an **inverted U** — base at the distal urethra, limbs to the bladder neck or beyond, **widening proximally** to preserve vascularity at the distal lateral flap margins. Compared with the inverted T, the inverted U gives excellent lateral exposure at the mid-vagina and extends proximally as needed. Saline infiltration aids dissection.
4. **Vaginal flap**: raise the anterior vaginal wall flap in the plane between vaginal wall and periurethral fascia, with countertraction; **preserve the fascia and avoid entering the UD**.
5. **Fascia**: incise the periurethral fascia **transversely**; develop proximal and distal fascial flaps without entering the UD.

6. **Excision:** grasp the UD and dissect it back to its origin within the fascial leaves, opening it if necessary; identify the ostium; **remove the entire mucosalised wall**, including small adherent or inflamed portions of urethral wall at the ostium, so that no mucosa remains to cause leakage or recurrence. Historical aids (urinary or Fogarty catheters in the UD, gauze packing, methylene blue, silicone or cryoprecipitate casts) are usually unnecessary.
7. **Urethral closure:** the catheter is often visible after excision. Close over a catheter as small as **12 Fr without long-term stricture risk — watertight, tension-free, absorbable**. A circumferential UD may require **segmental urethral resection** and complex reconstruction.
8. **Fascial closure:** reapproximate the periurethral fascial flaps **perpendicular to the urethral suture line** to minimise overlap and fistula risk, **obliterating all dead space**.
9. **Optional Martius flap:** a fibrofatty labial flap over the fascia as an extra layer. Indications are not agreed, but it is favoured with **poor-quality tissue, attenuated fascia or marked inflammation** to reduce wound breakdown and urethrovaginal fistula; harvest complications are low ($\leq 2\%$) with good cosmesis.
10. **Vaginal closure:** advance and reapproximate the vaginal flap with running interlocking absorbable suture — a **three-layer closure (four with a Martius flap)** — and place an antibiotic-impregnated vaginal pack.
 - **Pseudodiverticulum** (fascia attenuated or absent): reconstruct with an interpositional **Martius flap** or a graft such as a pubovaginal fascial sling. Preserving and reconstructing the fascia is paramount to prevent recurrence, close dead space and avoid fistula (Fig. 122.32).
 - **Rare anterior UD without a posterior component: suprameatal, infrapubic incision**, same principles. Surgery in the **prone-jackknife** position has also been described, reflecting the ergonomic difficulty of transvaginal surgery.

Postoperative care

Antibiotics for **24 hours** → remove vaginal pack → discharge with closed catheter drainage; antispasmodics liberally for bladder spasm → **pericatheter VCUG at 14–21 days** → no extravasation: remove catheters; extravasation: **repeat weekly** until resolved — most resolve within several weeks conservatively.

Complications

TABLE 122.2 COMPLICATIONS OF TRANSVAGINAL URETHRAL DIVERTICULECTOMY

Complication	Range of reported incidence
Urinary incontinence	1.7–16.1%
Urethrovaginal fistula	0.9–8.3%
Urethral stricture	0–5.2%
Recurrent urethral diverticula	10–22%
Recurrent urinary tract infection	0–31.3%
Hypospadias or distal urethral necrosis; bladder or ureteric injury; vaginal scarring or narrowing (dyspareunia)	Not available

Modified from Dmochowski R: Surgery for vesicovaginal fistula, urethrovaginal fistula, and urethral diverticulum. In Walsh PC, Retik AB, Vaughan ED Jr, et al., editors: Campbell's urology, ed 8, 2002.

BODY TEXT VERSUS TABLE 122.2 FOR THE SAME ENDPOINTS

Endpoint	Table 122.2	Body text	Reason for difference
Incontinence	1.7–16.1%	De novo SUI 7–16%; up to about half if minor leakage counted	Different definitions (any incontinence vs significant SUI vs any new SUI)

Endpoint	Table 122.2	Body text	Reason for difference
Recurrent UD	10–22%	Long-term series 10–22%; pooled meta-analysis 9.8%	Range of individual series vs pooled estimate

- Careful adherence to the principles minimises complications. In a national surgical quality database, **30-day complications were few and minor, with UTI commonest.**
- **Common overall: recurrent UTI, urinary incontinence and recurrent UD.**
- One small series linked **large (>4 cm), lateral or horseshoe UD** to more complications; another found similar rates except for **more UTI after complex repairs.**

URETHROVAGINAL FISTULA AFTER DIVERTICULECTOMY: DISTAL VERSUS PROXIMAL

Distal to the sphincteric mechanism	Proximal (bladder neck, or mid-urethra with incompetent bladder neck)
Few symptoms beyond a split stream or vaginal voiding	Considerable symptomatic leakage
Asymptomatic → may not need repair (some request it)	Repair , considering a Martius flap ; timing relative to the index operation is controversial

- **Fistula prevention:** meticulous technique, good haemostasis, avoiding infection, **preserving the periurethral fascia**, a well-vascularised vaginal flap and a **multilayered closure with nonoverlapping suture lines.**

Persistence or recurrence of symptoms after urethral diverticulectomy

- Obstructive and irritative LUTS may resolve, but **symptoms persist or recur in a substantial number regardless of UD complexity, approach or technical success.** Symptoms caused by surgery may settle with time; otherwise investigate for **recurrent UD, a new UD or urethral stricture.**
- Excision of a **recurrent UD** is less likely to resolve preoperative symptoms.
- **Causes of recurrence:** incomplete removal, inadequate urethral closure, residual dead space, other technical factors. **Long-term recurrence about 10–22%;** pooled contemporary rate about 10%.
- **Risk factors:** **circumferential configuration** (independently associated on multivariate analysis), **complex configuration** (simple UD rarely recur after one operation; complex ones often need several), **proximal location** and **multiple UD.** Horseshoe UD had low recurrence in one series.
- **Reoperation is difficult** — altered anatomy, scarring, obscured planes — and **fistula and recurrence are commoner in reoperative cases;** secondary procedures are less successful at complete removal.

KEY POINTS: MANAGEMENT OF FEMALE URETHRAL DIVERTICULA

- Investigate concomitant incontinence thoroughly before surgery; strongly consider treating bothersome SUI at the time of diverticulectomy.
- Transvaginal excision with urethral reconstruction is the commonest approach.

BLADDER DIVERTICULA VERSUS FEMALE URETHRAL DIVERTICULA

Feature	Bladder diverticulum	Female urethral diverticulum
Origin	Mucosal herniation through detrusor (congenital weakness or high voiding pressure)	Infected, obstructed periurethral gland rupturing into the lumen
Driving cause to treat	BOO or neurogenic dysfunction	Infection and ostial obstruction; coexisting SUI
Commonest cancer	UCC (SCC second)	Adenocarcinoma (UCC, then SCC)
Key diagnostic study	Cystoscopy with cytology; VCUG for emptying	MRI ; cystoscopy weak for ostium

Feature	Bladder diverticulum	Female urethral diverticulum
Operate for	Complications, not size	Symptoms or complications
Commonest operation	Diverticulectomy with outlet treatment	Transvaginal excision and layered reconstruction

Concept — *Every principle of urethral diverticulectomy — complete sac removal, watertight closure, fascia preserved and closed off-axis, dead space obliterated — exists to prevent its three failures: recurrence, fistula and incontinence.*